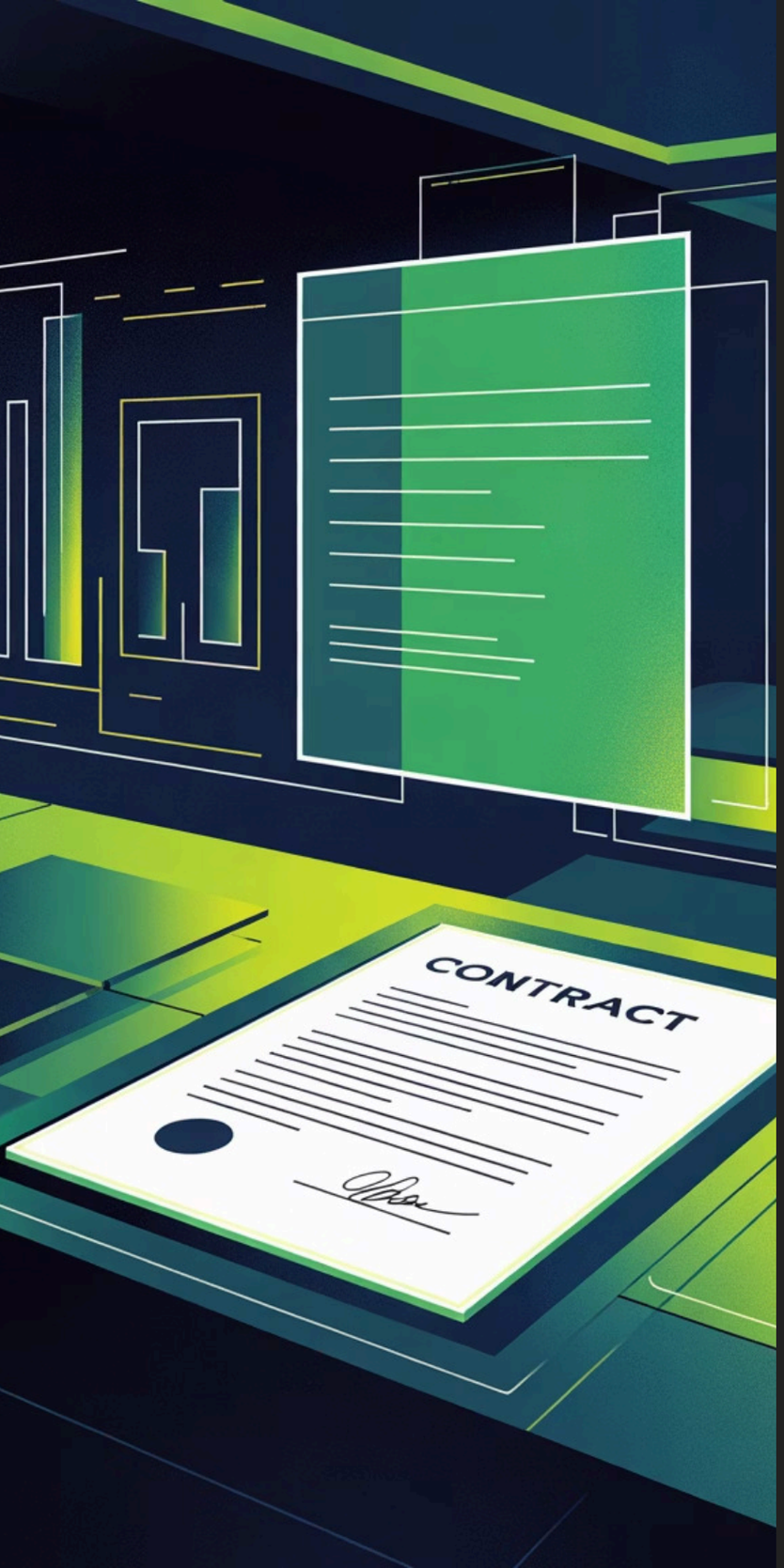

BEYOND 512 TOKENS

Taming Legal Jargon with Specialized NLP

An Automated System for Legal Document Extraction, Classification, Summarization, and Simplification

Presented to: Legal Technologists & NLP Researchers

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The Solution: Specialized Tools for a Specialized Job

We move beyond general-purpose models by integrating two state-of-the-art architectures designed to overcome the domain and length challenges.

Domain-Specific Mastery: LegalBERT

Pre-trained from scratch on a massive 12 GB corpus of diverse legal texts (contracts, legislation, court cases) to ensure deep understanding of "legalese."

Long-Document Handling: Longformer

Designed to overcome the 512-token limit. It uses an efficient 'sparse attention' mechanism to process inputs up to 16,000+ tokens, allowing it to read the entire document at once.

The integration of these two models provides both the legal context awareness and the necessary technical capacity for scaling to real-world legal documents.

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System Architecture: A Four-Stage NLP Pipeline

Our proposed system validates key information using established academic benchmarks and datasets.

Information Extraction (IE)

Identify and highlight specific data points (e.g., "Governing Law," "Parties," "Effective Date").

Methodology: Extractive tasks validated using the **CUAD** (Contract Understanding Atticus Dataset) framework.

Clause Classification

Automatically categorize every sentence or clause (e.g., "Liability," "Termination," "Indemnification").

Methodology: Classification techniques benchmarked against **LexGLUE** and **ContractNLI** datasets.

Summarization (Abstractive)

Generate a short, comprehensive summary (a "tl;dr") of a multi-page document.

Methodology: Utilizing Longformer-Encoder-Decoder (**LED**) models, optimized for long document summarization.

Simplification

Translate complex, technical text into clear, plain English accessible to a layperson.

Methodology: Generative models fine-tuned on limited simplification datasets, addressing a key research gap.

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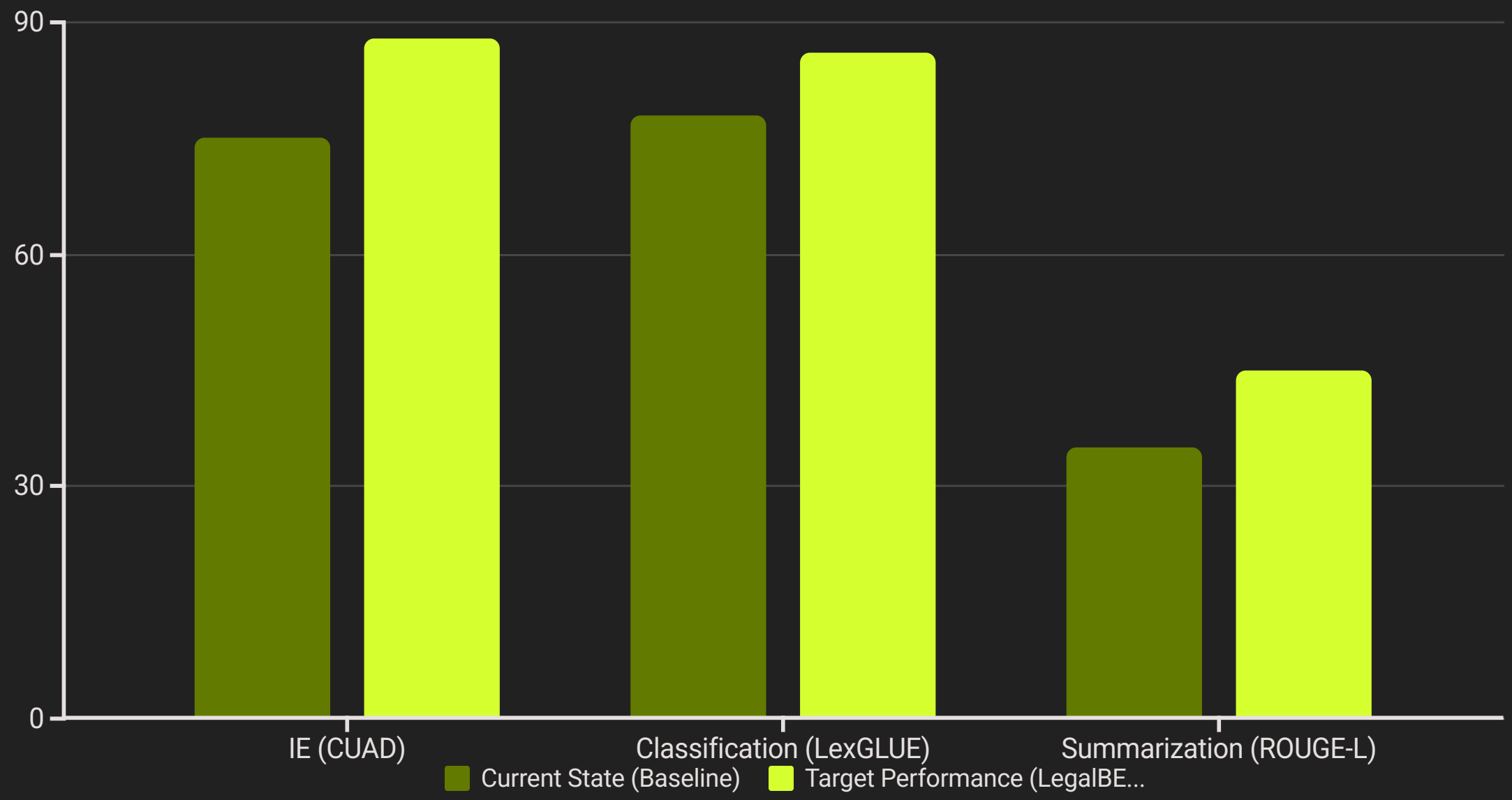
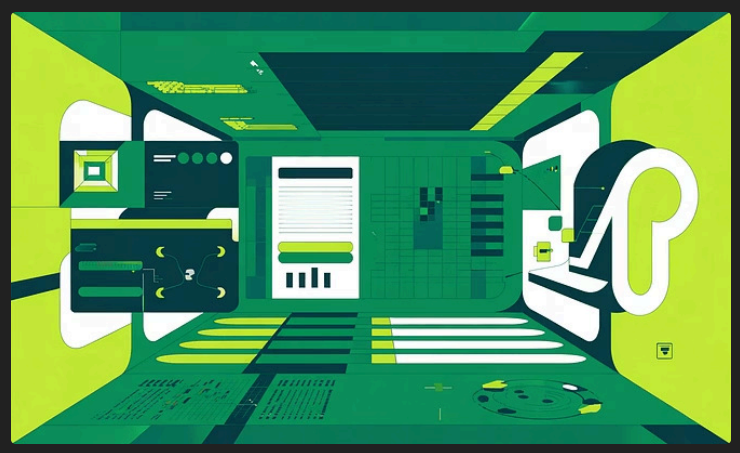
Key Metrics: Performance Targets (Targeting 85% F1 Score)



The integration of LegalBERT for domain knowledge and Longformer for context length is projected to yield significant performance gains across all core tasks, moving us past baseline models.



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THE IMPACT

Transforming Legal Productivity and Accessibility

Accelerated Due Diligence

Cut down contract review time from hours to minutes, enabling faster deal closure and compliance checks.

Automated Risk Flagging

Identify and flag non-standard, unusual, or high-risk clauses instantly, minimizing exposure.

Cost Reduction

Lower operational costs for law firms and corporate legal departments by automating low-level review tasks.

Enhanced Public Accessibility

Empower laypeople and small businesses to understand the documents they sign without relying on expensive legal counsel.

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Impact for Legal Professionals: Streamlining CLM

This system directly enhances Contract Lifecycle Management (CLM) by automating key, labor-intensive stages.

1

Intake & Extraction

Auto-populate metadata fields and key terms into CLM systems immediately upon document upload.

2

Risk Analysis

Flag deviations from organizational standards or preferred clause libraries during the negotiation phase.

3

Post-Execution Search

Enable highly granular search functionality across millions of documents based on categorized clauses, not just keywords.

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Impact for the Public: Leveling the Playing Field

By simplifying complex legal documents, we address issues of social inequality and comprehension barriers.

1 Breaking the Knowledge Barrier

Making rental agreements, terms of service, and loan documents truly comprehensible for the average citizen.

2 Reducing Information Asymmetry

Providing a free, instant tool to check document risk and key obligations, reducing the need for expensive preliminary consultations.

3 Empowerment

Giving individuals the confidence and knowledge to negotiate terms or understand their rights before signing a binding document.

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Conclusion: Integrating Specialized AI for Real-World Impact

This project successfully addresses the twin challenges of domain specificity (legalese) and document length inherent in legal text processing by integrating the best-in-class models.

Proof of Concept Validated

We have a working framework based on established datasets for extraction, classification, and long-document summarization.

Overcoming Technical Limits

The combination of LegalBERT and Longformer bypasses the critical 512-token constraint for production deployment.

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Future Work & Research Expansion

Multilingual Expansion

Extend the models to handle documents from civil law jurisdictions and common law systems in other major languages (e.g., French, German, Spanish).

Legal Question Answering (LQA)

Move beyond simplification to complex LQA: "What happens if I break this contract?" requiring deeper inference and reasoning.

Reasoning & Outcome Prediction

Explore sophisticated models for predicting legal outcomes based on factual inputs and case law analysis.

Thank you. I am now happy to take your questions.

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